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Paper Code: TMC 401

Paper Name COMPUTER GRAPHICS AND VISUAL COMPUTING

Time: 3 Hours

MM: 100

Each question has three parts (a,b,c) . Attempt any two parts of each question

Q1.

(10 X 2 = 20 Marks)

- Explain parallel Projection? How perspective projection is different from parallel projection? What is the importance of vanishing point in projection?
- Calculate the coordinates of a given unit cube having a point A, at center (0,0,0) rotated about z axis by 90 degree anti clock wise. Show the transformation of the rotated cube.
- What do you understand by back face detection. Explain Z-buffer algorithm.

Q2.

(10 X 2 = 20 Marks)

- What is ROTATIONAL transformation? Explain types of transformations.
- For a given triangle of coordinates A (10,30) B(30,30) C(20,15) find the following
 - Reflection about X-axis
 - Scaling about point A
- Explain Bresenham circle drawing algorithm? Why circle is divided in octants in circle drawing algorithms?

Q3.

(10 X 2 = 20 Marks)

- Digitize the pixel points using DDA algorithm for a line segment A(20,12) B(31,19) using DDA line drawing algorithms
- Write short note on any two
 - Shearing Transformation
 - Oblique Projection
 - Principle of animation
- What is inside-outside test? Explain odd-even parity and winding number rule for checking if a point is inside or outside a given area.

Q4.

(10 X 2 = 20 Marks)

- Derive bresenham line algorithm? What are the limitations of bresenham algorithm.
- Explain 4-connected and 8-connected model for filling. Explain boundary fill algorithm.
- Explain the working of LCD and LED. How raster scan is different from random scan?

Q5.

(10 X 2 = 20 Marks)

- Describe the Sutherland Hodgeman polygon clipping algorithm. What are convex and concave polygons?
- Consider two raster systems with the resolutions of 640x480, 1280x1024. What size frame buffer (In KB) is needed for each of these systems to store 12 bits/pixel , if 10 seconds video with 30 fps is loaded.
- Briefly explain the Cohen-Sutherland line clipping algorithm?